

ECE253/CSE208 Introduction to Information Theory

Lecture 3: Shannon Information and Entropy

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- Chap 2 of *Elements of Information Theory (2nd Edition)* by Cover–Thomas
- Chap 7 of *Information Theory: From Coding to Learning* by Polyanskiy–Wu
- *Introduction to Learning & Decision Trees*, CMU

Information is Surprise

Shannon Information

Shannon Entropy

Boltzmann Entropy

Implications of Entropy

Label Invariance

Joint and Conditional Entropy

Chain Rule of Entropy

KL Divergence (Relative Entropy)

f -Divergence

Conditional KL

Cross Entropy

- Many ways to communicate: phone call, text msg, email, Morse code, etc.
- Shannon thought of them as ways of producing strings of symbols.
- How to measure the amount of information contained in such a string?
- Shannon's idea: **information is related to what's new; related to surprise.**
- Think about reading news, words like “the” or “a” are pretty unsurprising (redundant since we understand the message without them). The real essence of the message lies in words that aren't as common, such as “alien” or “invasion”.
- This suggests a measure of surprise for each word: the frequency with which it appears in the English language.
- More generally, **the information of an outcome is inversely proportional to its probability.**

Shannon's Self-information: A Measure of Surprise

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The amount of information generated at the source X by the occurrence of the outcome x is defined as

$$I(x) := \log \frac{1}{p(x)}$$

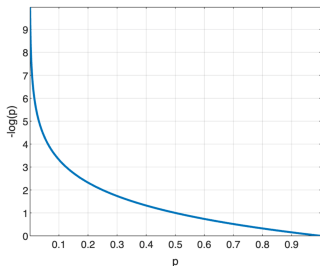


Figure: The less probable an event is, the more surprising it is and has more information.

Why logarithm

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- It is a monotonically increasing transformation
- It converts the value of information into the number of bits
- Must-happen thing carries no information: $p(x) = 1 \implies I(x) = 0$
- For independent RVs X and Y , the self-information of a joint outcome is additive:

$$I(x, y) = I(x) + I(y).$$

Shannon Entropy: A Central Role in Information Theory

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Entropy is a **measure of uncertainty** of a random variable, which is defined as the average Shannon information $\mathbb{E}_X[I(X)]$.

Definition (Entropy)

$$H(X) := \mathbb{E}_X[I(X)] = \mathbb{E}_X \left[\log \frac{1}{p(X)} \right] = - \sum_{x \in \mathcal{X}} p(x) \log p(x).$$

- $p(x) = 0 \implies 0 \log 0 = 0$ (by convention or justified by the L'Hopital's rule).
- Entropy is always nonnegative: $H(X) \geq 0$.
- Units for different log bases: $\log_2 \rightarrow \text{bits}$; $\log_e \rightarrow \text{nats}$; $\log_{10} \rightarrow \text{harleys/dits/bans}$
- $H_b(X) = (\log_b a) H_a(X) \Leftarrow$ the change of base formula: $\log_b x = \frac{\log_a x}{\log_a b}$.

Axiomatic Definition of Entropy

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Entropy should satisfy the following properties:

Continuity [$H(\mathbf{p})$ is continuous in $\mathbf{p}$]: The amount of info associated with an outcome should vary continuously as the probability of that outcome changes.

Maximum [$H(0.5) = 1$]: The amount of info associated with a set of outcomes cannot be increased if those outcomes are equally probable.

Grouping [$H(p_1, p_2, \dots, p_m) = H(p_1 + p_2, p_3, \dots, p_m) + (p_1 + p_2)H\left(\frac{p_1}{p_1 + p_2}, \frac{p_2}{p_1 + p_2}\right)$]: The amount of info associated with a set of outcomes is obtained by adding the info of each outcome.

It can be shown that $H(p_1, p_2, \dots, p_m) = -\sum_{i=1}^m p_i \log p_i$ is the only function satisfying the above properties; see page 53 of Cover-Thomas's book.

See more properties of $H(X)$: <http://www.mtm.ufsc.br/~taneja/book/node6.html>

Grouping Property of Entropy

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Interpretation

- **First stage (coarse uncertainty):** which group the outcome belongs to
- **Second stage (refined uncertainty):** which element within a group
- Total entropy = uncertainty of the group + expected uncertainty within the group

Example

Let $(p_1, p_2, p_3) = (0.2, 0.3, 0.5)$. Group $\{1, 2\}$:

$$H(0.2, 0.3, 0.5) = H(0.5, 0.5) + 0.5 H(0.4, 0.6).$$

Name and Notation

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- Shannon: “Von Neumann told me, ‘You should call it entropy, for two reasons. In the first place your uncertainty function has been used in statistical mechanics under that name, so it already has a name. In the second place, and more important, nobody knows what entropy really is, so in a debate you will always have the advantage.’”
- The term entropy was coined in 1865 by the German physicist Rudolf Clausius from Greek word “en-” = in + “trope” = change or transformation.
- H is used because Shannon chose it to mirror thermodynamic entropy, and the notation became canonical through widespread adoption.
- Second law of thermodynamics: Entropy of an isolated system increases over time. An isolated system is one that exchanges neither energy nor matter with its surroundings. The system's total energy, volume, and particle number are fixed.

Boltzmann Entropy

- Boltzmann entropy is a thermodynamic property of a system, which is a bridge between the microscopic world (positions and momenta of all the atoms) and the macroscopic (e.g., total energy E , volume V , pressure P , temperature T , etc).
- Boltzmann entropy of a system is defined as

$$S_B = \kappa_B \ln \Omega \text{ (J/K)}$$

where Ω is the number of microstates consistent with a given macrostate, and $\kappa_B \approx 1.38 \times 10^{-23}$ J/K is the Boltzmann constant.

- Entropy is determined by the number of random microstates. It's related to the amount of additional information needed to specify the exact physical state.
- Entropy and Energy: Low entropy = energy concentrated; High entropy = energy spreads out.

Example: Particles Distributed Among Energy Levels

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Consider a system of n particles and m distinct energy levels $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$.

Let n_i denote the number of particles occupying energy level ε_i , with $\sum_{i=1}^m n_i = n$.

- A macrostate is specified by the occupation numbers $(n_1, n_2, \dots, n_m)$.
- Permuting particles among the same energy level does not change the physical configuration. Hence, the $n_i!$ permutations within each energy level do not generate new microstates and *must be counted only once*.
- Therefore, the number of distinct microstates corresponding to the macrostate $(n_1, \dots, n_m)$ is $\Omega = \frac{n!}{n_1! n_2! \dots n_m!}$ (multinomial coefficient).
- Applying Stirling's approximation (good for large integers n),

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \implies \ln n! \approx n \ln n - n,$$

we obtain

$$\ln \Omega \approx n \ln n - \sum_{i=1}^m n_i \ln n_i$$

Link Boltzmann Entropy to Shannon Entropy

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Define the empirical probabilities $p_i = \frac{n_i}{n}$, $i = 1, \dots, m$. Then, $\ln \Omega \approx -n \sum_{i=1}^m p_i \ln p_i$.

Substituting into Boltzmann's entropy formula, we obtain the **Gibbs entropy**

$$S_G \approx k_B n \left(- \sum_{i=1}^m p_i \ln p_i \right).$$

This expression is identical to Shannon entropy, up to constants and the choice of log-base.

1. Boltzmann: macrostate-based, microstate counting vs Gibbs: distribution-based, ensemble entropy.
2. Boltzmann: equilibrium-focused vs Gibbs: valid out of equilibrium.

Entropy as a Measure of Average Information

How much is the average information of a given RV X ?

Example (Bernoulli distribution)

Consider $X \sim \text{Bern}(p)$, we have $H(X) = -p \log p - (1-p) \log(1-p) =: f(p)$.

Special cases: $H(0.5) = 1$ bit; $H(0) = H(1) = 0$.

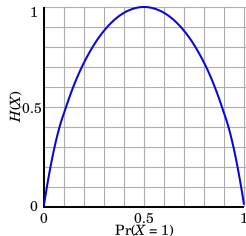


Figure: $H(X)$ achieves its maximum when $p = \frac{1}{2}$; i.e., X has the greatest uncertainty when both outcomes are equally likely.

Implication 1: Entropy for Equally Probable Outcomes (1/2)

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We can use 2 bits ($\{00, 01, 10, 11\}$) to fully describe a system of 4 equally probable states. In general, $\log_2(m)$ bits are needed for m equally probable states.

- For uniform distribution random variable $X \in \mathcal{X} \implies H(X) = \log(|\mathcal{X}|)$.
- Uniform probability yields maximum entropy (maximum uncertainty).
- For equiprobable events, the entropy increases with the number of outcomes $\implies$ Hence, casting a die has higher entropy than tossing a coin.

Implication 1: Entropy for Equally Probable Outcomes (2/2)

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Example (An unfair coin)

Suppose we have a two-sided coin where $P(X_h) = 0.9, P(X_t) = 0.1$.

$$H(X) = 0.469 \text{ bits} \implies m = 2^{H(X)} = 1.38 \text{ equally probable outcomes}$$

A coin with $H(X) = 0.469$ bits has the same entropy as a dice with 1.38 sides. ■

Insight:

A random variable with entropy $H(X)$ bits provides enough Shannon information to choose $m = 2^{H(X)}$ equally probable outcomes.

Implication 2: Entropy for Average Description Length (1/2)

Example (Variable-Length Encoding)

Consider a random source taking symbols from the set: $\{A, B, C, D\}$.

Case 1: Uniform distribution with $P = 0.25$ for each letter

$$A \rightarrow 00, B \rightarrow 01, C \rightarrow 10, D \rightarrow 11 \Rightarrow \bar{L} = 2 \text{ bits/symbol}$$

Case 2: Non-uniform distribution with $P = (0.70, 0.26, 0.02, 0.02)$

$$A \rightarrow 0, B \rightarrow 10, C \rightarrow 110, D \rightarrow 111$$

Average length: $\bar{L} = 0.7 \times 1 + 0.26 \times 2 + 0.02 \times 3 + 0.02 \times 3 = 1.34 \text{ bits/symbol}$

Entropy: $H = 1.09 \text{ bits}$

Observation: $\bar{L} > H(P)$



Implication 2: Entropy for Average Description Length (2/2)

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The above example reflects the key idea of Huffman encoding: More likelihood symbols should be assigned with shorter length of codes to minimize the average length of the description.

Insight:

Entropy $H(X)$ provides a lower bound on the average length of the shortest description of the random variable X . That is, entropy can be viewed as the cost of encoding the specific distribution X .

Source Coding Theorem

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Theorem (Shannon's Source Coding Theorem)

For any discrete random variable X with finite alphabet and any uniquely decodable code $C : \mathcal{X} \rightarrow \mathcal{A}^$, the expected codeword length satisfies*

$$\mathbb{E}[\ell(X)] \geq H(X).$$

Label Invariance of Entropy

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Key Property: Label (Relabeling) Invariance

Entropy depends only on the PMF, not on the numerical values taken by the RV.

$$H(X) = - \sum_x p(x) \log p(x)$$

Invariance statement: For any one-to-one mapping f ,

$$H(X) = H(f(X)).$$

- Outcomes can be relabeled without changing entropy
- Entropy measures *uncertainty of outcomes*, not their magnitudes
- Only the probability weights matter

Entropy vs. Variance

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Do not be confused:

- entropy (uncertainty of outcomes),
- variance (numerical spread),
- unpredictability (difficulty of guessing).

Examples

- Let $P(X = 0) = 1 - \epsilon$ and $P(X = \frac{1}{\epsilon}) = \epsilon$. Then $\text{Var}(X) = \frac{1}{\epsilon} - 1$ (large), while $H(X) = H(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$.
- $Y \in \{1, 2, 3, 4\}$ with $P(Y = i) = \frac{1}{4}$: smaller spread, higher entropy ($H(Y) = 2$ bits).

From Discrete Entropy to Differential Entropy

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- **Discrete entropy** measures uncertainty of outcomes:

$$H(X) = - \sum_x p(x) \log p(x) \quad (\text{bits})$$

- **Differential entropy** extends the idea to continuous variables:

$$h(X) = - \int p(x) \log p(x) dx$$

- Key differences:
 - $h(X)$ can be *negative*
 - $h(X)$ depends on measurement units and scaling
 - $h(X)$ is *not* invariant under reparameterization
 - $h(X)$ measures “uncertainty per unit volume,” so changing coordinates changes the volume — but not the information.

Joint and Conditional Entropy

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Cross Entropy

Consider a pair of random variables (X, Y) with joint distribution $p(x, y)$.

Joint entropy:

$$H(X, Y) = \mathbb{E}_{(X, Y)} \left(\log \frac{1}{p(X, Y)} \right) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y).$$

Conditional entropy:

$$\begin{aligned} H(Y|X) &= \mathbb{E}_{(X, Y)} \left(\log \frac{1}{p(Y|X)} \right) = - \sum_{x, y} p(x, y) \log p(y|x) \\ &= - \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log p(y|x) \\ &= \sum_{x \in \mathcal{X}} p(x) H(Y|X = x) \end{aligned}$$

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- $H(Y|X = x)$ is the specific conditional entropy, i.e., the entropy of Y knowing that X takes a specific value x . $H(Y|X)$ is the expectation of $H(Y|X = x)$.
- $H(f(X)|X) = 0$ for any function $f(\cdot) \Leftarrow$ no randomness of $f(X)$ if X is given.
- If $X \perp Y \implies H(X|Y) = H(X)$
- $H(X|Y) \neq H(Y|X)$ in general: Let $X \sim \text{Bern}(\frac{1}{2})$ and $Y \equiv 0$. Then $H(Y | X) = 0$ while $H(X | Y) = H(X) = 1$.
- *Preview:* Mutual information measures how much knowing one variable reduces uncertainty about another, which is defined as

$$I(X; Y) = H(X) - H(X|Y)$$

Chain Rule of Joint Entropy (1/2)

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Chain rule: $H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$.

Proof: Take expectation on both sides of $\log p(X, Y) = \log p(X) + \log p(Y|X)$.

Interpretation:

- Joint uncertainty of (X, Y) is equal to uncertainty of X plus uncertainty of Y after knowing X .
- $H(Y|X)$ measures how much information, on average, X encodes about Y .
- Now if we first learn the value of X , we have gained $H(X)$ bits of information. Once X is known, we only need $H(Y|X) = H(X, Y) - H(X)$ additional bits to describe the state of the whole system.

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Theorem (Chain rule for entropy)

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, X_{i-2}, \dots, X_1).$$

Proof.

$$H(X_1, X_2) = H(X_1) + H(X_2 | X_1)$$

$$H(X_1, X_2, X_3) = H(X_1) + H(X_2 | X_1) + H(X_3 | X_2, X_1)$$

$$\vdots$$

$$H(X_1, X_2, \dots, X_n) = H(X_1) + H(X_2 | X_1) + \dots + H(X_n | X_{n-1}, \dots, X_1)$$



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Corollary

- *Bayes' rule for conditional entropy: $H(Y|X) = H(X|Y) - H(X) + H(Y)$*
- $H(X, Y|Z) = H(X|Z) + H(Y|X, Z)$
- $H(f(X)) \leq H(X)$ for any function $f(\cdot) \rightarrow$ *post-processing reduces entropy*

Example of Joint and Conditional Entropy (1/2)

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Example

Consider two RVs X and Y , whose joint distribution is given below. Find $H(X)$, $H(Y)$, $H(X|Y)$, $H(Y|X)$, and $H(X, Y)$.

$Y \setminus X$	1	2	3	4	$p(y)$
1	1/8	1/16	1/32	1/32	1/4
2	1/16	1/8	1/32	1/32	1/4
3	1/16	1/16	1/16	1/16	1/4
4	1/4	0	0	0	1/4
$p(x)$	1/2	1/4	1/8	1/8	

Example of Joint and Conditional Entropy (2/2)

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$$H(X) = H\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right) = \frac{1}{2} \log 2 + \frac{1}{4} \log 4 + 2 \times \frac{1}{8} \log 8 = \frac{7}{4} \text{ bits}$$

$$H(Y) = H\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right) = 2 \text{ bits}$$

$$H(X|Y) = \sum_y p(y) H(X|Y=y)$$

$$= \frac{1}{4} \times \left(H\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right) + H\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}\right) + H\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right) + H(1, 0, 0, 0) \right) = \frac{11}{8} \text{ bits}$$

Similarly, we can get $H(Y|X) = \frac{13}{8}$ bits, $H(X, Y) = \frac{27}{8}$ bits.

Relative Entropy (Kullback-Leibler Divergence)

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Definition (Relative Entropy)

The relative entropy (a.k.a. KL divergence) between two probability massive functions $p(x)$ and $q(x)$ over **the same sample space** is defined as:

$$D_{\text{KL}}(P\|Q) = \mathbb{E}_p \left(\log \frac{p(X)}{q(X)} \right) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)}$$

- If $p(x) = 0$, the term $p(x) \log \frac{p(x)}{q(x)}$ is defined to be zero. If there exists x such that $p(x) > 0$ and $q(x) = 0$, then $D(p\|q) = \infty$.
- $\log \frac{p(x)}{q(x)}$ is the log likelihood ratio (LLR). Hence, KL divergence is the expected LLR.

Interpretation of KL Divergence

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- In many applications, p represents the data-generating (true) distribution, while q represents a model or estimated distribution.
- In Bayesian inference, $D_{\text{KL}}(P||Q)$ is a measure of the information gained when one revises the belief from the prior distribution q to the posterior distribution p . In other words, it's the amount of info lost when q is used to approximate p .
- $D_{\text{KL}}(P||Q)$ is the average number of extra bits needed to encode the data, due to the fact that we used distribution q to encode the data instead of the true distribution p .

KL Divergence as Information Gain (Bayesian Example)

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- **Unknown parameter:** Let $\theta = \Pr(\text{Heads})$ denote the (unknown) bias of a coin, which is modeled as random variable.
- **Prior belief (very confident):** $q(\theta) = \mathcal{N}(0.5, 0.001)$.
- **Observed data:** 10 coin flips with 8 Heads and 2 Tails.
- **Posterior belief:** $p(\theta) = \mathcal{N}(0.8, 1)$, indicating a substantial shift in belief away from the prior.
- **Information gained from data:** $D_{\text{KL}}(p(\theta) \parallel q(\theta)) \approx 781$ bits.
- **Interpretation:**
 1. If you encode draws of θ using a code optimized for the prior instead of the posterior, you waste about 781 extra bits per draw on average.
 2. Revising from the prior to the posterior requires about *781 extra bits* to encode θ using a prior-based code, quantifying how strongly the data contradicted the prior belief.

KL Divergence Is Not a Metric

$D_{\text{KL}}(P\|Q)$ is **NOT** a distance metric: it is asymmetric and violates the triangle inequality.

$$D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P), \quad D_{\text{KL}}(P\|Q) \not\leq D_{\text{KL}}(P\|R) + D_{\text{KL}}(R\|Q)$$

Example

Consider two distributions $p(x)$ and $q(x)$, where $x \in \mathcal{X} = \{0, 1\}$. Let $p(0) = 1 - r$ and $p(1) = r$ while $q(0) = 1 - s$ and $q(1) = s$. Then,

$$D_{\text{KL}}(P\|Q) = (1 - r) \log \frac{1 - r}{1 - s} + r \log \frac{r}{s}$$

$$D_{\text{KL}}(Q\|P) = (1 - s) \log \frac{1 - s}{1 - r} + s \log \frac{s}{r}$$

If $r = \frac{1}{2}$ and $s = \frac{1}{4}$, then $D_{\text{KL}}(P\|Q) = 0.2075$ bits and $D_{\text{KL}}(Q\|P) = 0.1887$ bits. ■

KL Divergence Is Directional: KL vs Reverse KL

Let P be the true distribution and Q an approximation.

	Forward KL $D_{\text{KL}}(P\ Q)$	Reverse KL $D_{\text{KL}}(Q\ P)$
Expectation under	P	Q
$P(x) > 0, Q(x) = 0$	∞	finite
$Q(x) > 0, P(x) = 0$	finite	∞
Typical behavior	mode-covering	mode-seeking
Common use	MLE	Variational inference

- **Mode-covering:** Penalizes assigning zero or small probability to any region where P has mass. As a result, the optimal Q spreads probability mass to cover all modes of P , often overestimating variance.
- **Mode-seeking:** Penalizes assigning probability mass to regions where P is small or zero. As a result, the optimal Q concentrates on one high-density mode of P , often underestimating uncertainty.

Minimize Forward KL

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- Fix the true distribution P and optimize over a model Q_θ

$$\min_{\theta} D_{\text{KL}}(P \| Q_\theta)$$

- Equivalent form:

$$D_{\text{KL}}(P \| Q_\theta) = \text{constant} - \mathbb{E}_{x \sim P}[\log Q_\theta(x)]$$

- Thus, minimizing forward KL is equivalent to MLE (Maximum Likelihood Estimation):

$$\max_{\theta} \mathbb{E}_{x \sim P}[\log Q_\theta(x)]$$

- Consequences:

- Mode-covering behavior
- Penalizes missing any mass of P
- Standard in density estimation, supervised learning, and generative modeling

Minimize Reverse KL

- Fix the target distribution P (often a posterior)

$$\min_{\phi} D_{\text{KL}}(Q_{\phi} \| P)$$

- Reverse KL expands as

$$D_{\text{KL}}(Q_{\phi} \| P) = \mathbb{E}_{z \sim Q_{\phi}} [\log Q_{\phi}(z) - \log P(z)]$$

- Practical motivation:

- Expectations under P are intractable; under Q_{ϕ} are tractable.
- Leads to the ELBO (evidence lower bound) formulation

- Consequences:

- Mode-seeking behavior
- Penalizes placing mass where P is small
- Common in mean-field variational inference (MFVI)

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f -Divergence: A Unifying Family of Distribution Similarities

- **f -divergence:** For two distributions P and Q on the same space,

$$D_f(P\|Q) = \mathbb{E}_q \left[f \left(\frac{p(X)}{q(X)} \right) \right],$$

where $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ is convex and $f(1) = 0$.

- *Important special cases*
 - Forward KL: $f(t) = t \log t$ and Reverse KL: $f(t) = -\log t$
 - Jensen–Shannon divergence, total variation distance, Hellinger.
- *Applications in machine learning*
 - Many commonly used loss functions can be interpreted as instances of f -divergences.
 - Different choices of the function f impose different inductive biases (i.e., model assumptions).
 - Variational objectives and Generative Adversarial Networks (GANs) can be derived from f -divergence minimization.

Jensen–Shannon Divergence (Symmetrized KL)

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Define the mixture distribution

$$M = \frac{1}{2}(P + Q).$$

Jensen–Shannon (JS) divergence is defined as

$$D_{\text{JS}}(P\|Q) = \frac{1}{2} D_{\text{KL}}(P\|M) + \frac{1}{2} D_{\text{KL}}(Q\|M)$$

As an *f*-divergence:

$$f(t) = t \log t - (t + 1) \log\left(\frac{t+1}{2}\right)$$

- Symmetric: $D_{\text{JS}}(P\|Q) = D_{\text{JS}}(Q\|P)$
- Always finite, even with disjoint supports
- Bounded: $0 \leq D_{\text{JS}}(P\|Q) \leq 1$
- $\sqrt{D_{\text{JS}}}$ is a metric
- Used in the original GAN objective

Conditional Relative Entropy

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Definition (Conditional relative entropy)

Let $p(x, y)$ and $q(x, y)$ be joint PMFs with conditional distributions $p(y | x)$ and $q(y | x)$. The conditional KL divergence is defined as

$$D_{\text{KL}}(p(y|x) \parallel q(y|x)) = E_{p(x,y)} \left(\log \frac{p(Y|X)}{q(Y|X)} \right) = \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log \frac{p(y|x)}{q(y|x)}.$$

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Theorem (Chain rule for relative entropy)

$$D_{\text{KL}}(p(x, y) \parallel q(x, y)) = D_{\text{KL}}(p(x) \parallel q(x)) + D_{\text{KL}}(p(y|x) \parallel q(y|x))$$

Corollary (KL divergence is additive for independent distributions)

Let

$$p(x, y) = p_1(x) p_2(y), \quad q(x, y) = q_1(x) q_2(y),$$

i.e., both joint distributions factorize into independent marginals. Then

$$D_{\text{KL}}(p(x, y) \parallel q(x, y)) = D_{\text{KL}}(p_1(x) \parallel q_1(x)) + D_{\text{KL}}(p_2(y) \parallel q_2(y)).$$

Proof of Chain Rule of Relative Entropy

Proof.

$$\begin{aligned} D_{\text{KL}}(p(x, y) \parallel q(x, y)) &= \sum_{x, y} p(x, y) \log \frac{p(x, y)}{q(x, y)} \\ &= \sum_{x, y} p(x, y) \log \frac{p(x)p(y|x)}{q(x)q(y|x)} \\ &= \sum_{x, y} p(x, y) \log \frac{p(x)}{q(x)} + \sum_{x, y} p(x, y) \log \frac{p(y|x)}{q(y|x)} \\ &= D_{\text{KL}}(p(x) \parallel q(x)) + D_{\text{KL}}(p(y|x) \parallel q(y|x)) \end{aligned}$$



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- **Definition:** The cross entropy between distributions P and Q is defined as

$$H(P, Q) \triangleq -\mathbb{E}_{X \sim P} [\log Q(X)] = H(P) + D_{\text{KL}}(P \| Q),$$

where $H(P) = -\mathbb{E}_{X \sim P} [\log P(X)]$.

- **Asymmetry:** In general,

$$H(P, Q) \neq H(Q, P).$$

- **Optimization equivalence:** For fixed P ,

$$\arg \min_Q H(P, Q) = \arg \min_Q D_{\text{KL}}(P \| Q).$$

- **Coding interpretation:** $H(P, Q)$ measures the average code length required to encode samples drawn from the true distribution P using an optimal code designed for Q .

Application of Cross Entropy In Machine Learning

Classification task: Identify to which of a set of classes a new observation belongs, on the basis of a training set of data containing observations whose class membership is known. For the training set $\{\mathbf{x}_i, y_i\}_{i=1}^N$, $\mathbf{x}_i$ is the feature vector (a.k.a. input or explanatory variables, regressors, covariates, predictors), and y_i is the true label of sample i .

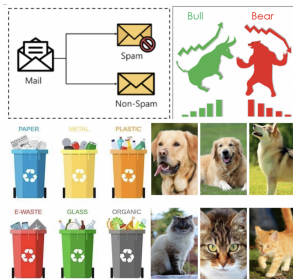


Figure: Examples of classification.

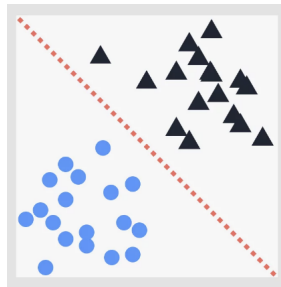


Figure: Binary classification.

Cross-Entropy Loss for Binary Classification

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- Let $y_i \in \{0, 1\}$ denote the observed label of sample i , and assume **deterministic labels** (no label noise).
- A model outputs a predicted probability via logistic regression (sigmoid function):

$$\hat{y}_i \triangleq \Pr(Y = 1 \mid X = \mathbf{x}_i) = \sigma(\mathbf{w}^\top \mathbf{x}_i) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}_i}}, \quad \Pr(Y = 0 \mid X = \mathbf{x}_i) = 1 - \hat{y}_i.$$

- Define
 - P : a (degenerate) Bernoulli distribution with parameter y_i ; i.e.,

$$\Pr(Y = 1 \mid X = \mathbf{x}_i) = y_i,$$

- Q : a Bernoulli distribution with parameter $\hat{y}_i$.
- The per-sample cross entropy between P and Q is

$$H(P, Q) = -y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i),$$

which is the standard **binary cross-entropy (log-loss)**.

- The training objective is obtained by averaging $H(P, Q)$ over all samples.

Cross Entropy Loss (a.k.a. log-loss)

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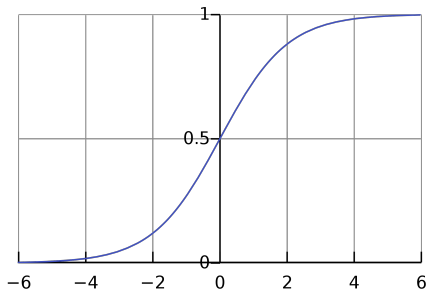


Figure: Logistic function: Maps real-valued inputs to probabilities in $(0, 1)$, with saturation near 0 and 1 and highest sensitivity around the decision boundary at zero.

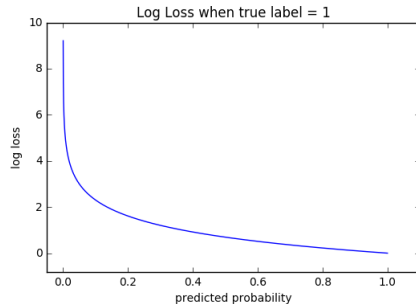


Figure: Cross entropy loss: $-\log(\text{probability assigned to the true label})$; confident correct predictions incur small loss, while confident incorrect predictions incur large loss, strongly penalizing overconfident errors.

Information-Theoretic Intuition

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- In information theory, the *surprise* of observing label y under a model Q is

$$-\log Q(y \mid \mathbf{x}).$$

- The cross entropy between the true distribution P and the model Q is the *expected surprise*:

$$H(P, Q) = \mathbb{E}_{(x,y) \sim P} [-\log Q(y \mid \mathbf{x})].$$

- Minimizing cross entropy therefore makes the true labels, *on average*, less surprising under the model.

Statistical Interpretation

- **Logistic regression assumption:**

$$Y \mid X = \mathbf{x} \sim \text{Bern}(\hat{y}(\mathbf{x})), \quad \hat{y}(\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}).$$

- Thus, for one sample $(\mathbf{x}, y)$, the **model likelihood (PMF)**:

$$Q_{\mathbf{w}}(y \mid \mathbf{x}) = (\hat{y}(\mathbf{x}))^y (1 - \hat{y}(\mathbf{x}))^{1-y}.$$

- **Negative log-likelihood** of one sample:

$$-\log Q_{\mathbf{w}}(y \mid \mathbf{x}) = -y \log \hat{y}(\mathbf{x}) - (1 - y) \log(1 - \hat{y}(\mathbf{x})),$$

which is exactly the **binary cross-entropy loss**.

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Why Cross-Entropy Loss?

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- **Proper scoring rule:** The expected loss is minimized when the predicted probabilities match the true conditional probabilities.
- **Convexity:** For logistic regression, the cross-entropy loss is convex in the model parameters, enabling efficient optimization.
- **Smooth and differentiable:** Provides informative gradients everywhere, unlike the 0–1 loss.
- **Probabilistic interpretation:** Directly corresponds to maximum likelihood estimation and has a clear information-theoretic meaning.

Takeaway of Cross Entropy

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Key Insight

Cross entropy is the natural loss for logistic regression because it measures how well the predicted probabilities explain the observed labels.

- Minimizing cross entropy = maximizing likelihood.
- Penalizes wrong predictions in proportion to confidence.
- Connects information theory, statistics, and optimization.